

Algorithms and Complexity

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Module 1

Automata Theory

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Lesson 1.0 Introduction

- Course structure and navigation

Reading • 15 min
- How to learn effectively on this course

Reading • 15 min
- Introduction to the course

Video • 55 sec
- Course Syllabus

Reading • 10 min

- Lesson 1.1 Finite automata
- Introduction

Video • 1 min
- Basic definitions, letters, strings

Video • 4 min
- What is an automaton?

Video • 8 min
- Basics

Practice Assignment • 30 min

- Lesson 1.2 Deterministic automata
- Finite automata – example (part 1)

Video • 3 min
- Finite automata – example (part 2)

Video • 8 min
- Working with Automata Simulator

Video • 6 min
- Design an automata to accept a simple language using Automata Simulator

Dialogue • 30 min
- Language of the automata

Video • 4 min
- Recognise a language

Video • 4 min
- Exercises with hints and tips

Reading • 1h

- Lesson 1.3 Non-deterministic Automata
- Deterministic finite automata (DFA) vs nondeterministic finite automata (NFA)

Video • 4 min
- DFA example

Video • 3 min
- DFA example in Automata Simulator

Video • 4 min
- DFA Simulator

Lab • 20 min
- Computation by NFA

Video • 3 min
- NFA example

Video • 5 min
- NFA example in Automata Simulator

Video • 2 min
- NFA Simulator

Lab • 20 min
- Exercises with hints and tips

Reading • 10 min

- Lesson 1.4 Summary and Assessments
- Conclusion

Video • 3 min
- Check your understanding: End of module 1

Graded Assignment • Grade: 100%

Module 2

Searching and Sorting Algorithms

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Lesson 2.1 Introducing Algorithms

- What is an algorithm ?

Video • 4 min
- Tell us about an algorithm that you use

Discussion Prompt • 20 min
- Representing algorithms

Video • 8 min
- Simple algorithms – insertion sort

Video • 5 min
- How else would you sort?

Practice Assignment • 30 min
- Simple algorithms – bubble sort

Video • 5 min

- Lesson 2.2 Sorting and Search Techniques
- Selection sort

Video • 4 min
- Insertion, bubble and selection sort

Practice Assignment • 25 min
- When we should use which sorting algorithm?

Dialogue • 15 min
- Sorting algorithms simulator: Bubble Sort and Insertion Sort

Lab • 20 min
- Exercises with hints and tips

Reading • 10 min
- Search techniques

Video • 4 min
- Binary and sequential search

Practice Assignment • 12 min

- Lesson 2.3 Heap sort
- Binary trees and heaps

Video • 3 min
- Heapify algorithm

Video • 4 min
- Heap sort

Video • 6 min
- Binary trees and heaps

Reading • 1h
- Heap sort

Practice Assignment • 20 min

- Lesson 2.4: Summary and Assessments
- Conclusion

Video • 1 min
- Check your understanding: End of module 2

Graded Assignment • Grade: 100%

Module 3

Recursive and Iterative Algorithms

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Lesson 3.1 Recursive and Iterative Algorithms

- Recursion

Video • 7 min
- Iterative algorithms

Video • 4 min
- Recursive algorithms example

Video • 5 min
- Recursive and iterative algorithms

Practice Assignment • 25 min
- Quick sort

Video • 6 min
- Quick sort example

Video • 4 min
- Quick sort

Practice Assignment • 20 min
- Partition and quick sort algorithms

Reading • 1h 25m
- Lesson 1 exercises with hints and tips

Reading • 10 min

- Lesson 3.2 Merging
- Merging lists

Video • 4 min
- How does this magic work?

Practice Assignment • 30 min
- Merge sort

Video • 4 min

— Karen, machine

Exercises with hints and tips

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Now it's time to put into practice the concepts that you have learnt so far in this module. Attempt the following exercises. If you get stuck, you can refer to the hints and tips at the end of this page.

Please note that these exercises are optional, further practice – but we strongly recommend that you engage with them in order to test your knowledge and see where you might need to do additional study.

1. Using merge sort, sort the following list showing all of the steps:
A: 13, 1,16, 8, 2, 9, 4, 5.

2. In most programming languages, you do not need to implement any sorting algorithms as they provide you with some built-in sorting functions. Try to find out which algorithm this programming language uses. Start with Java or Python.

3. Assume $S = \{ \text{Nkem (N), Elena (E), Fatima (F)} \}$ and $H = \{ \text{Whittington (W), Royal Free (R), Highgate (H)} \}$. The list of the preferences is as follows. Find the stable match using the Gale-Shapley algorithm. Show the steps of the algorithms clearly.

Name of nurse	Order of preference of hospital
NKem	H R W
Elena	W R H
Fatima	W H R

Name of hospital	Order of preference of nurse
Whittington	N F E
Royal Free	F N E
Highgate	F E N

4. Solve the problem again with following list of preferences:

Name of nurse	Order of preference of hospital
NKem	H W R
Elena	W H R
Fatima	W R H

Name of hospital	Order of preference of nurse
Whittington	N E F
Royal Free	F E N
Highgate	F N E

Hints and tips for each question:

- Start by halving the list until each sublist has only one element. Then merge them until there is one list.
- Search the web based on the programming language to find the answer.
- First, all the hospitals make an offer to their first choice of nurse. Then each nurse decides which hospital to choose. Each hospital that their preferred choice of nurse rejected makes an offer to their second choice. This process continues until all the nurses have accepted an offer.
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